∴ The remaining $(m-2)$ elements of A must be mapped within the remaining $(m-1)$ elements of B apart from $f(a)$.

So, there will exist two distinct elements b_1 and b_2 in B s.t. $f(a_1)=b_1$ and $f(a_1)=b_2$ for some $a_1 \in A$. This violates the defn. of function given in Defn. 1.

∴ f must be one-to-one.

∴ f is one-to-one if and only if it is onto. □

75) Prove or disprove each of these statements about the floor and ceiling functions.

a) $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ for all real numbers x.

Soln: By Defn. 12, $\lfloor x \rfloor$ is the largest integer smaller than or equal to x.
By Defn. 12, $\lceil \lfloor x \rfloor \rceil$ is the smallest integer larger than or equal to $\lfloor x \rfloor$
∵ $\lfloor x \rfloor$ is an integer, ∴ $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ □

b) $\lfloor 2x \rfloor = 2\lfloor x \rfloor$ whenever x is a real number.

Soln: $\lfloor 2\times 0.1 \rfloor = \lfloor 0.2 \rfloor = 1$ $\quad 2\lfloor 0.1 \rfloor = 2\cdot 1 = 2$
∴ $\lfloor 2x \rfloor \neq 2\lfloor x \rfloor$ whenever x is a real number. □

c) $\lceil x \rceil + \lceil y \rceil - \lceil x+y \rceil = 0$ or 1 whenever x and y are real numbers.

Soln: Let, $\lceil x \rceil = a$, $\lceil y \rceil = b$ and $\lceil x+y \rceil = c$. By Defn. 12, $a,b,c \in \mathbb{Z}$.
By Property (1b), $a-1 < x \leq a$, $b-1 < y \leq b$ and $c-1 < (x+y) \leq c$
By Property (1d), $x \leq a < x+1$, $y \leq b < y+1$, $x+y \leq c < (x+y)+1$

$x+y \leq a+b < (x+y)+2$ $\quad x+y \leq c < (x+y)+1$ $\quad$ XX Wrong approach.

∴ $a = x+\varepsilon_1$, $b = y+\varepsilon_2$, $c = (x+y)+\varepsilon_3$ where $0 \leq \varepsilon_1, \varepsilon_2, \varepsilon_3 < 1$.
$a+b = (x+y)+(\varepsilon_1+\varepsilon_2)$ $\quad$ ∴ $a+b-c = (\varepsilon_1+\varepsilon_2)-\varepsilon_3$.
$0 \leq \varepsilon_1 < 1$ $\quad 0 \leq \varepsilon_2 < 1$ $\quad 0 \leq \varepsilon_3 < 1$.
$0 \leq \varepsilon_1+\varepsilon_2 < 2$. $\quad$ ∴ $\varepsilon_1+\varepsilon_2-\varepsilon_3 < 2-0 \Rightarrow \varepsilon_1+\varepsilon_2-\varepsilon_3 < 2$.
∵ $a+b-c \in \mathbb{Z}$, ∴ $\varepsilon_1+\varepsilon_2-\varepsilon_3 \in \mathbb{Z}$. So, they can only take up the values of 0 or 1